

Course Name: VLSI Design Lab

Course Code: EEE426

OPEN ENDED LAB REPORT

Experiment Name: The objective of this experiment is to implement the Boolean function in DSCH and MICROWIND and verify their credibility with timing diagram.

Date: 9th December 2022 Thursday

Group: I

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- I. **Objective:** The objective of this experiment is to implement the Boolean function $Y = \overline{(A.B + B.C + D)}$ in DSCH and MICROWIND and verify their credibility with timing diagram.

II. **Theory:**

Boolean Logic

In Boolean algebra, the binary variables and the operations of logic are utilized. This form of algebraic expression, which is referred to as a Boolean Expression, is utilized in order to describe the Boolean Function. The constant values 1 and 0 make up the Boolean expression, together with the symbols for the logical operations and the binary variables.

Truth Table

A truth table is a mathematical table that is used in logic, more specifically in connection with Boolean algebra, Boolean functions, and propositional calculus. It is used to set out the functional values of logical expressions on each of their functional arguments, that is, for each combination of values taken by their logical variables. A truth table is also known as a truth matrix. Truth tables may be utilized to demonstrate whether a propositional statement is true for all proper input values, or in other words, whether it is logically sound.

Complementary Metal-Oxide Semiconductor (CMOS) Design

Pull-Up and Pull-Down Networks: A Pull-Up Network (PUN) and a Pull-Down Network (PDN) are both part of a complimentary MOS gate, which is a combination of two networks (PDN). In CMOS design, all inputs are routed to both the PUN and the PDN. The purpose of PUN is to establish a connection between V_{DD} and V_{out} in order to draw V_{out} to the logic value of "1," whereas the purpose of PDN is to establish a connection between GND and V_{out} in order to pull V_{out} to the logic value of "0." Both the PUN and the PDN have distinct but complementary functions.

In a typical configuration, the PDN will be made up of NMOS devices, and the PUN will be made up of PMOS devices.

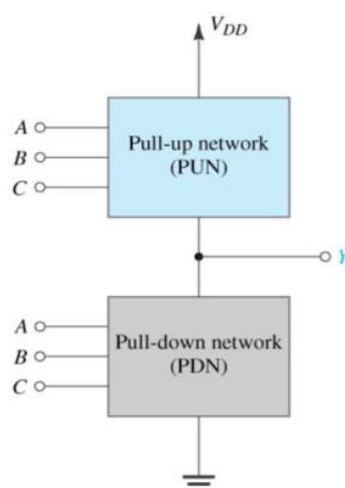


Figure 1 CMOS Design using PDN and PUN

The fact that NMOS transistors create "strong zeros" and PMOS devices generate "strong ones" is the primary rationale for combining these two types of devices. Pulling the output all the way up to logic high is required for PUN (i.e. V_{DD}). If we are going to utilize an NMOS transistor, then the voltage level of Y will be Grounded. Where V_{DD} is the threshold voltage of the NMOS transistor, and V_{DD} is the voltage level of Y if we use a PMOS transistor instead. Therefore, PMOS generates a strong '1,' but NMOS generates a weak '1.'

Stick Diagram

A stick diagram is a draft of the actual layout; it provides an abstract picture that bridges the gap between the schematic and the layout. It demonstrates the relative positioning of the components, which is helpful in communicating the layer information using color codes. The notation for stick diagrams may be seen down below. They are also able to be shown using a line art or shades of gray technique.

P diff	
N diff	
Poly	
Metal	
N-Well	
Contact	

Rules to be followed for drawing stick diagrams are:

- A contact is made whenever two adjacent parts of the same layer cross each other.
- If the separate layers just cross one another, then there is no contact established between them; however, if there must be contact between them, then it must be expressed expressly in the shape of a dark shaded circle.
- An NMOS enhancement transistor is formed if a polysilicon cross n-type of diffusion takes place; an NMOS depletion type transistor is formed if a polysilicon cross n-type of diffusion takes place in conjunction with an implant; and a PMOS enhancement transistor takes place if a polysilicon cross n-type of diffusion takes place on its own.
- In CMOS design style, PMOS devices must be positioned in the draw up portion (above the demarcation line), while NMOS devices must be positioned in the pull-down part (below the demarcation line).
- A demarcation line is utilized in between p-type and n-type devices in order to clearly identify between the two types of devices.
- A cross mark is required to be placed on both the V_{DD} and V_{SS} rails in order to successfully produce an n-well and a p-well contact for PMOS and NMOS devices, respectively.

Layout Design Rules:

Layout design rules describe how small features can be and how closely they can be reliably packed in a manufacturing process. Industrial design rules are usually specified in microns. This makes migrating from one process to a more advanced process or a different foundry's process difficult because not all rules scale in the same way.

A Representation of the Layout That Is Layered:

The masks that are employed in CMOS are transformed into straightforward layout levels by the layer representation of layout. These levels are simpler and hence much simpler for designers to visualize. The following components are used as the foundation for CMOS design layouts:

- (1) Substrates or Wells: For NMOS devices, these wells are of the p type, whereas for PMOS devices, they are of the n type.
- (2) Diffusion areas: The formation of transistors takes place in these regions, which are also referred to as the active layer. NMOS transistors have an n+ definition, while PMOS transistors have a p+ definition.
- (3) Polysilicon layers: These are the layers that are utilized in the formation of the gate electrodes for the transistors.
- (4) Metal interconnects layers: These are the layers that are utilized to construct the power supply rails, the ground rails, the input rails, and the output rails respectively.
- (5) Contact and Via layers: These are the layers that are utilized to make the connections between the layers.

III. Procedure

For DSCH Shema

1. Run DSCH.exe
2. In order to build the circuit topology shown in figure 1, we referred to the "Symbol List" and employed the relevant logic gates and components.
3. In order to perform our logical function, we made use of two AND gates, one OR gate, and one inverter.
4. We sourced "Symbol List" for our use of the supply and ground symbols.
5. In this scenario, the button served as the input, and the LED served as the output.
6. While the simulation was running, we changed the input by pressing the button, and then we watched the state of the LEDs change. Red denotes logic 1, whereas black stands for logic 0. We terminated the simulation, and then we examined the timing diagram.
7. Afterward, utilizing Adobe Illustrator, a stick diagram showing the function that was provided was made.

For Microwind Layer

1. Run MICROWIND.exe.

2. The process technique with a thickness of 90 nm (0.9 m) was chosen by us from the following:
[File > Select Foundry > default.rul > open]
3. Following that, we drew four polysilicon layers, each with a width of 0.9 micrometers.
4. Over the polysilicon, a p+ diffusion layer that is at least 0.240 m (4 lambda) wide was drawn.
5. We began by drawing an n+ diffusion layer on top of the polysilicon that had a minimum width of 0.240 m (4 lambda). When a p+ diffusion layer is placed on top of polysilicon, a pMOS is produced, and when an n+ diffusion layer is placed on top of polysilicon, a nMOS is produced.
6. We drew an n-well layer that covered the pMOS by at least 0.36 micrometers (6 lambda) and was separated from the nMOS by at least 0.36 micrometers (6 lambda) (6 lambda).
7. Using metal layer, we make connections between the drain terminals of the transistors.
8. After that, we made a VDD rail by utilizing a metal layer of the same kind and attached it to the source of the pMOS.
9. In the same manner, we constructed a GND rail and attached it to the source of the nMOS.
10. We established various kinds of relationships in the appropriate locations.
11. In addition to the VDD rail and the n-well, we installed a "VDD supply."
12. On the ground rail, we affixed the letters "GND."
13. The polysilicon layers received an additional four clocks in the form of gate inputs A, B, C, and D.
14. A "visible node" was installed at the point where the metal drains connected. This is the Y terminal, which represents the output.
15. In order to determine whether there is a rule, we made use of ribbon's "Design Rule Checker."
16. Since there was no problem, we proceeded with the simulation and made sure that the timing diagram was accurate.

IV. Truth Table

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0

1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

V. Schematics

Schematic of CMOS Design from the provided Boolean Function

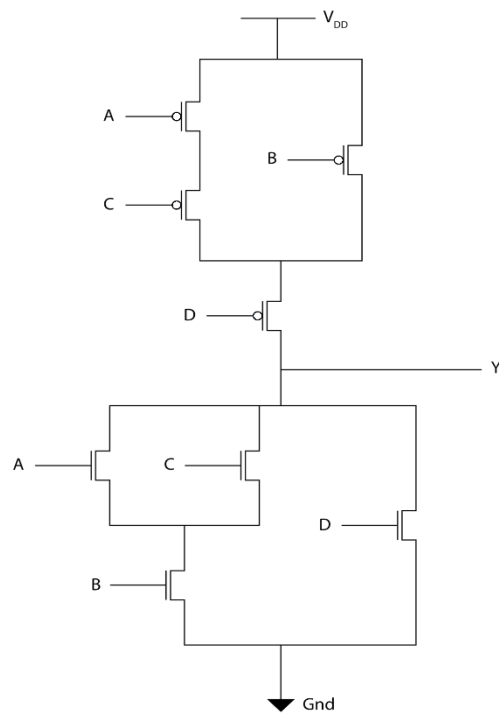
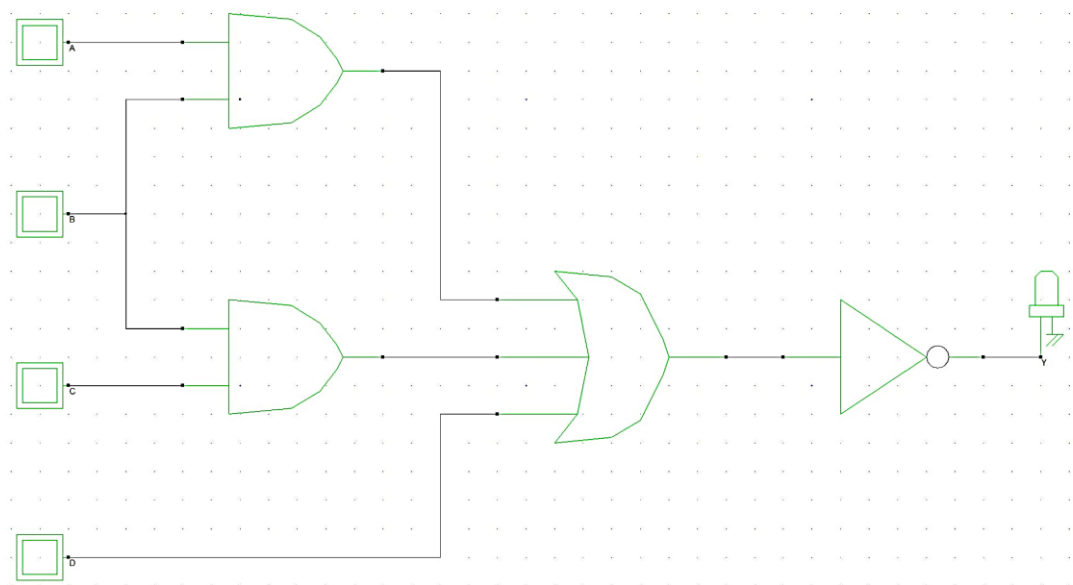


Figure 2 CMOS design of the given Boolean Function

Schematic of Logic Gate Design from the provided Boolean Function

Figure 3 Logic Gate Design from the provided Boolean Function



Stick Design from the provided Boolean Function

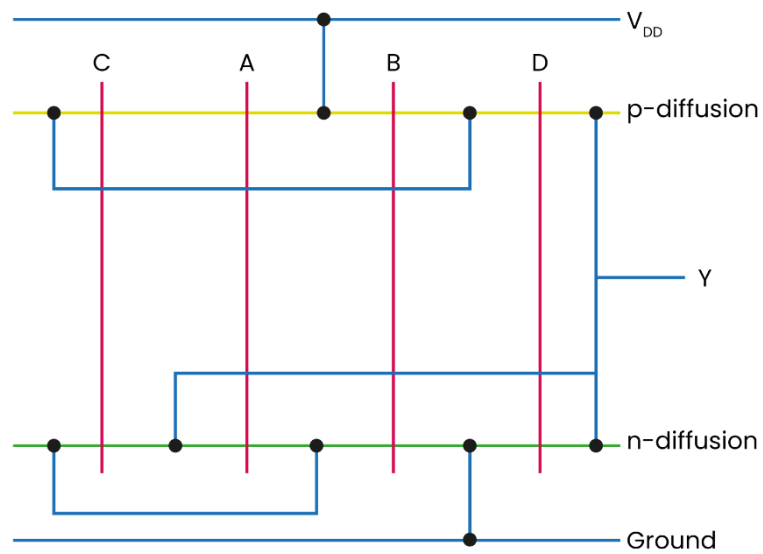


Figure 4 Stick Diagram of provided function

Schematic of MASK Design from the provided Boolean Function

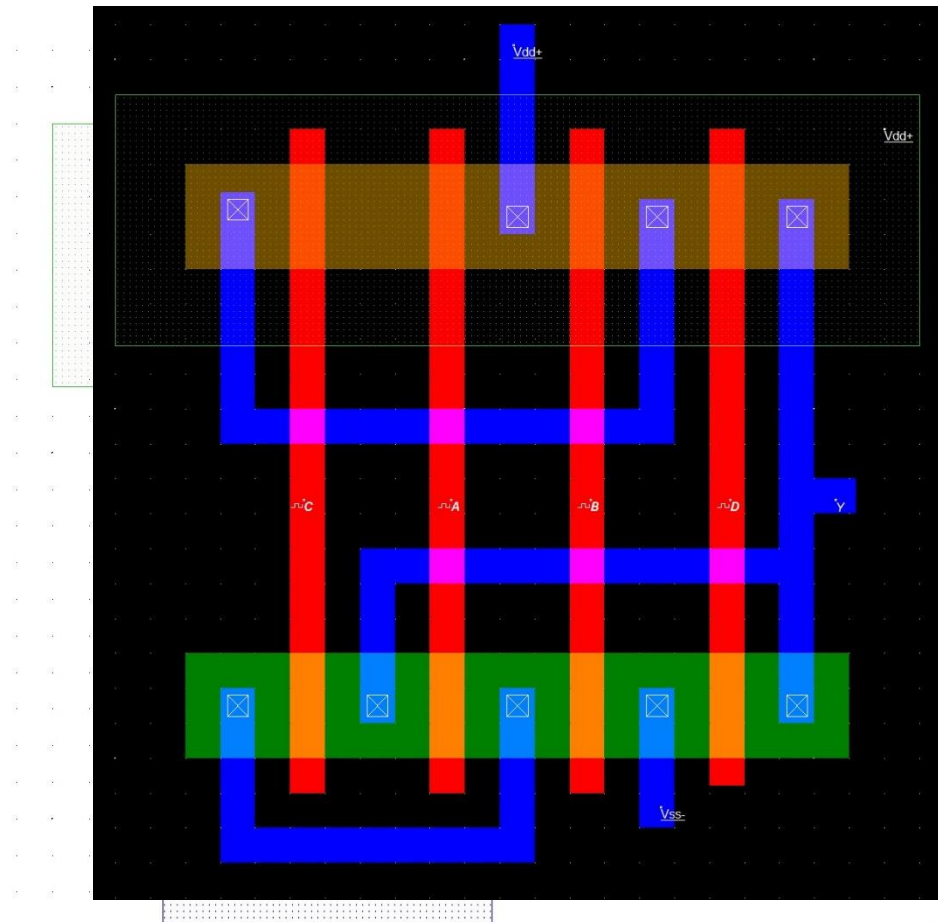


Figure 5 (a) MASK Design of the provided Boolean Function (b) Black background for better visibility

VI. Results & Analysis

Timing Diagram of the design schematic from Microwind

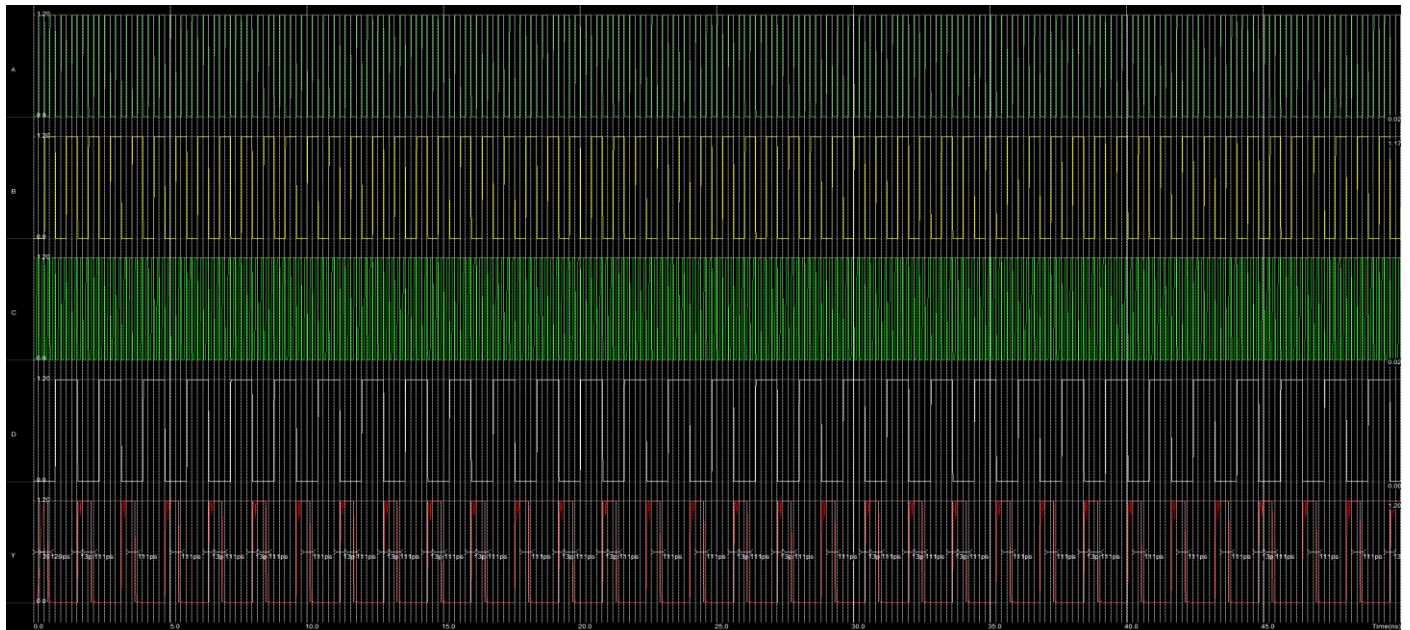


Figure 6: Timing Diagram with the time scale of 50ns

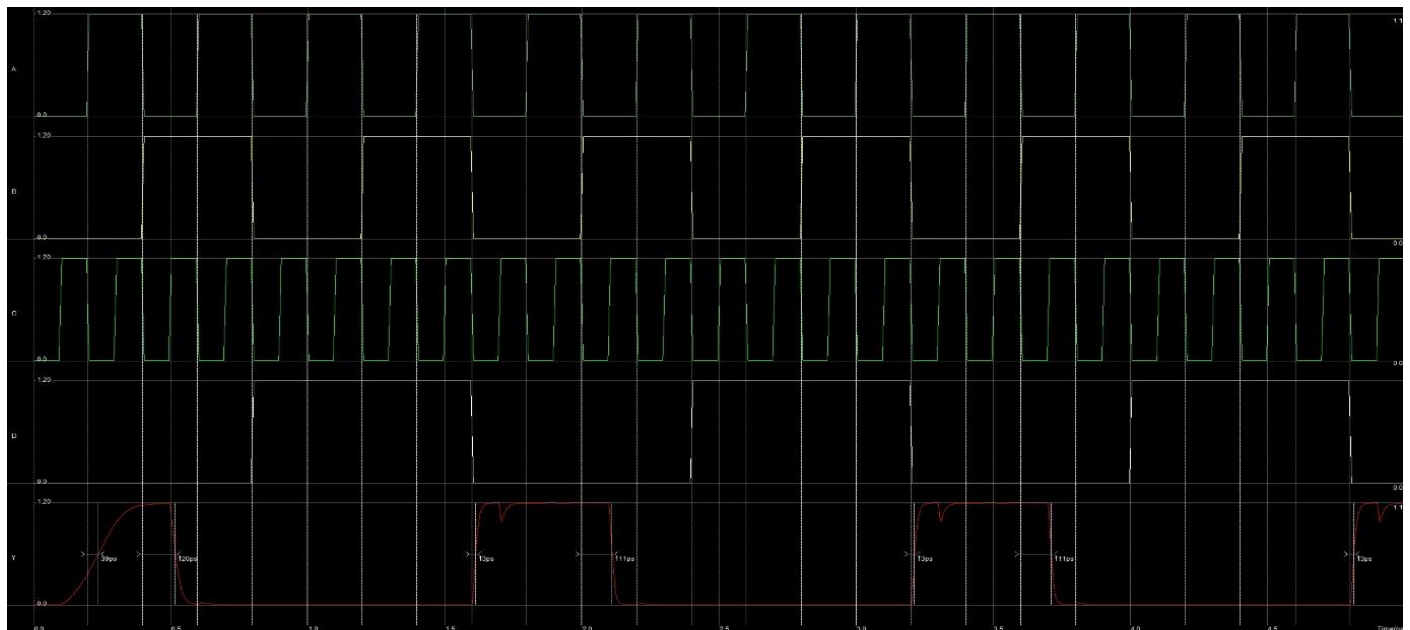


Figure 7: Timing Diagram with the time scale of 5ns

Timing Diagram of the design schematic from DSCH

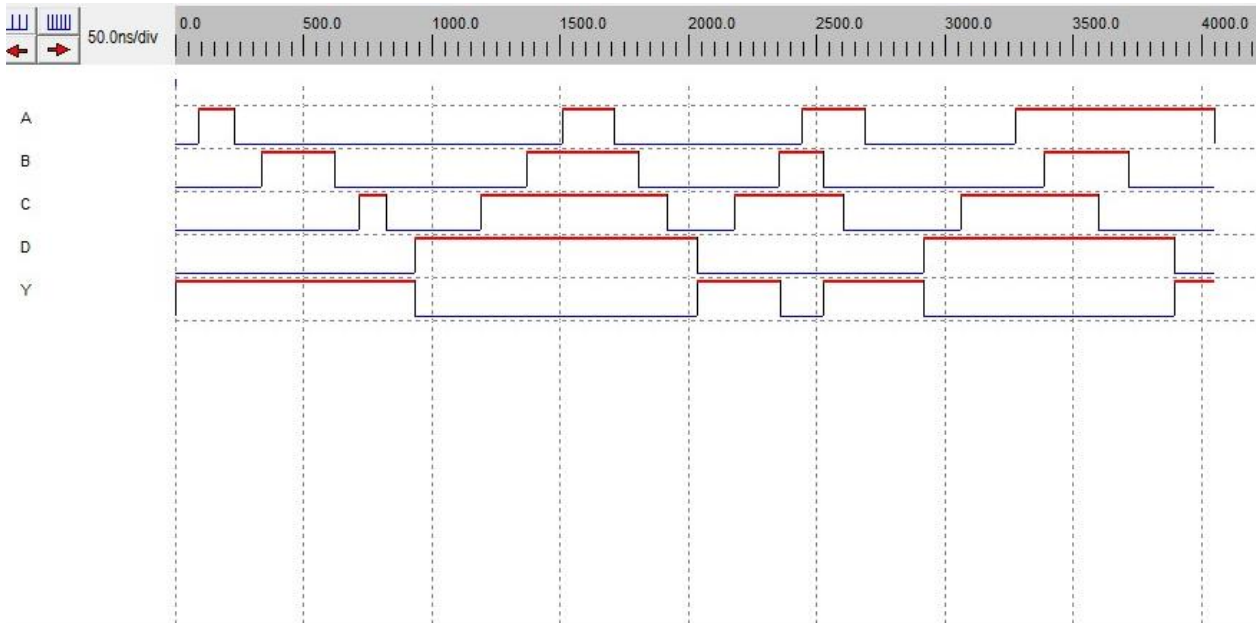


Figure 8: Timing Diagram with the time scale of 50ns/div

VII. Discussion

In the experiment we had to use previous knowledge and experiences from the prior lab reports. We need knowledge of CMOS Design, from the Boolean Function we translated it into a CMOS design using a Pull-Up Network and Pull-Down Network as shown in Fig. 2. The design of the stick diagram was developed completely based on the CMOS design, we located the N-MOS and P-MOS and the Drains and Sources of each MOSFET. The connections in between each MOSFETs were given with metal and a contact the illustration of the stick diagram in Fig. 4. This stick diagram was strictly followed for the MASK design in Microwind, the 90-nm (0.9-μm) process technology was adopted. Then we drew four 0.9-micrometer-wide polysilicon layers. A 0.240-μm (4-lambda)-wide p+ diffusion layer was drawn over polysilicon. We started by drawing a 0.240-μm-wide n+ diffusion layer on polysilicon (4 lambda). When a p+ diffusion layer is deposited on polysilicon, a pMOS is created. We drew an n-well layer that covered the pMOS by 0.36 micrometers (6 lambda) and separated it from the nMOS (6 lambda). Metal layer connects transistor drain terminals. Then, we constructed a VDD rail from the same metal layer and hooked it to the pMOS source. We built a GND rail and connected it to the nMOS. We formed proper contacts. We added a "VDD supply" to the VDD rail and n-well. We marked the ground rail "GND." Polysilicon layers A, B, C, and D got four more clocks. Where the metal drains met, a "visible node" was built. The output Y terminal. We used ribbon's "Design Rule Checker" to find rules. We ran the simulation and checked the timing diagram as there was no problem.

. For the design of schema in DSCH we translated the Boolean function give and used "AND" and "OR" logic gates. To construct the circuit topology depicted in Figure 1, we consulted the "Symbol List" and utilized the appropriate logic gates and components. To accomplish our logical function, we utilized two AND gates, one OR gate, and one inverter. We obtained "Symbol List" in order to utilize the supply and ground symbols. In this case, the button was the input and the LED were the output. We altered the input by pressing a button while the simulation was running, and then observed the state of the LEDs change. Red represents logic 1, whereas black represents logic 0. Following the conclusion of the simulation, we reviewed the timing diagram. Then, using Adobe Illustrator, a stick diagram depicting the provided function was created.

With both timing diagrams, Fig. 7 and Fig. 8, identical results can be seen, even though the Microwind timing diagram initially displays a lag indicating the time required for raising and lowering, but in DSCH software this occurs instantly.

It can observed in both the timing diagrams that when "A", "B", "C" and "D" all are off the output is High; "A", "B", "C" and "D" are "0", "0", "1" and "0" the output is High; "A", "B", "C" and "D" are "0", "1", "0" and "0" the output is High; "A", "B", "C" and "D" are "1", "0", "0" and "0" the output is High; "A", "B", "C" and "D" are "1", "0", "1" and "0" the output is High for all the other combinations of "A", "B", "C" and "D" the output is LOW.